

# Particulate Matter Variability in Urban and Rural Sites: Anthropogenic and Meteorological Drivers

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It is important to understand how location, whether urban or rural, or meteorological properties can affect air quality for general health. This report assesses particulate readings  $<2.5\mu\text{m}$  and  $<10\mu\text{m}$  (PM2.5 and PM10) in Rochester (rural) and Marylebone (urban) between January 2018 and February 2021. Part 1 assesses the differences in PM10 between urban and rural sites. Urban site measurements were found to be related to the work week commute, while rural site shows no corresponding diurnal pattern. The difference between sites is found to have a Mann-Whitney effect size of  $r=0.331$  (medium). Rochester PM10 shows a pronounced April peak, consistent with agricultural tillage resuspension. Part 2 assesses the impact of meteorological conditions on the measurements of PM at the rural site. A wind rose demonstrates that Easterly winds bring both PM2.5 and PM10 matter, while high pressure anticyclone trapping has links with soil-moisture based particulate dispersal. Meteorological conditions, particularly wind direction, pressure and precipitation are key drivers of PM variability alongside the urban/rural distinction.

## I. INTRODUCTION

PM is an important consideration for human health, as PM10/PM2.5 includes any particulate smaller than 10 or 2.5 micrometers respectively, which can lead to inhalation and deep penetration into the respiratory tract, where PM2.5 can reach the alveoli, so can result in serious impact to health, especially in vulnerable groups. [1]

Both urban and rural environments have crucial differences when comparing PM: In urban environments, traffic is a significant PM10 source through both exhaust emissions and mechanical resuspension of road dust [2]. The predictable structure of urban commuting further suggests that PM10 should exhibit a diurnal cycle, peaking during rush hours. This pattern will be absent from rural backgrounds where traffic is sparse, whereas counteracting this, rural areas do have heavy machinery which frequently disturbs soil and creates PM [3]

Physical processes also exist which link weather to PM. Not only does Easterly wind bring European PM to the UK [4], but low soil moisture can lead to increased PM emissions [3]. Furthermore, PM readings can be influenced by Anticyclones reducing winds and trapping PM at the observable layer [5]

This study therefore addresses two questions: whether PM10 differs systematically between urban and rural sites with a commuter-driven diurnal structure, and whether meteorological conditions - particularly wind direction, pressure, and precipitation - are significant drivers of PM variability at the rural site. It hypothesises that PM10 at the urban site will be elevated relative to the rural site, with a diurnal structure consistent with commuter traffic patterns, with the further hypothesis that both PM2.5 and PM10 will elevate under easterly winds, while high temperature and pressure, low wind speed, gust and precipitation will all be observed under elevated PM10 and PM2.5, both having very similar relationships to the factors that effect them.

## II. METHODOLOGY

The data this report uses consists of three datasets, two of these being for site-type air quality analysis and the third relating to weather data. The two datasets relating to air quality are sourced from DEFRA's UK-Air Archive, collected from Rochester and London Marylebone Road monitoring stations. The dataset stores PM10 and PM2.5 readings hourly between January 2018 and February 2021 and also contains status columns to verify whether that datapoint passed quality control. These readings are taken by Tapered Element Oscillating Microbalances with Filter Dynamic Measurement Systems (TEOM FDMS) to adjust for the volatility of the mass

The third dataset is a collection of readings from London Heathrow Airport, beginning from 1948 and contains meteorological data, which includes wind gust, wind speed, temperature, precipitation, air pressure and wind direction. For this analysis only the overlapping period between all 3 was used.

Despite quality control, random noise in TEOM FDMS sensors can produce negative readings at low particulate concentrations, which are physically impossible, so these were treated as measurement failures and thus replaced by NaN prior to analysis. Furthermore, readings flagged as Suspect (S) in the Status column for PM2.5 and PM10 were excluded by setting their corresponding values to NaN. Readings flagged as Not Verified (N) in the same columns were retained however given that the datasets had undergone prior cleaning.

To statistically test the difference between the observations at the urban and rural site, this report uses the Mann-Whitney U test, which is a non-parametric test for whether two independent samples are drawn from the same distribution. It was useful for this project, as it doesn't require the data to be normally distributed, the statistical significance is assessed via the p value, with effect size  $r$  calculated to quantify the magnitude of the

difference.

$$r = \frac{Z}{\sqrt{N}} \quad (1)$$

Where  $Z$  is the test statistic and  $N$  is the total number of observations.

Spearman Rank Correlation is also heavily used in the report, as it is a non-parametric measure of association between two variables ranging from -1 (perfect negative monotonic relationship) to 1 (perfect positive monotonic relationship). It was chosen rather than Pearson's as Pearson's assumes a linear relationship.

Binned means are used especially in Part 2, where bins of equal (or quartile) width are created, values put into them and the bin's mean PM is calculated, in an attempt to reveal trends obscured by scatter. Binned means are calculated with the formula

$$\bar{X}_k = \frac{1}{n_k} \sum_{i \in k} x_i \quad (2)$$

A wind rose is also used to accurately plot the relationship between PM observation and observed wind directions. A wind rose shows a sector in each compass direction and the radius of that sector is the size of the mean PM value recorded in that direction.

Data manipulation and analysis were performed in Python, using the pandas [6], numpy [7] and scipy [8] libraries. Visualisation was produced using matplotlib [9]

### III. RESULTS

#### A. Part 1: Urban vs Rural PM10

To establish whether a systematic difference exists in between sites, the mean PM10 was calculated for across the focus period. Fig. 1 shows that the urban Marylebone site recorded a mean PM10 of  $20.6 \mu\text{g}/\text{m}^3$  compared to  $14.4 \mu\text{g}/\text{m}^3$  at rural Rochester. A Mann-Whitney U test confirms this difference is statistically significant ( $p < 0.001$ ) with medium effect size ( $r = 0.331$ ), indicating that urban location is associated with meaningfully raised PM10, rather than just a detectable influence on PM10 concentration.

The bar charts remain insufficient for in depth analysis of the data however, due in part to their large standard deviations and majority overlap, which casts a shadow of doubt onto the conclusiveness of any findings.

To next ascertain whether the overall difference between the sites was consistent across seasons or whether seasonal patterns differed, the mean monthly PM10 was plotted for both of the sites across the studied period. Fig. 2 shows that Marylebone remains relatively stable throughout the year, with a modest April elevation superimposed on a consistently higher traffic baseline. Rochester by contrast shows a pronounced April peak

Mean PM10 Climatology Between Monitoring Sites (2018-2021)

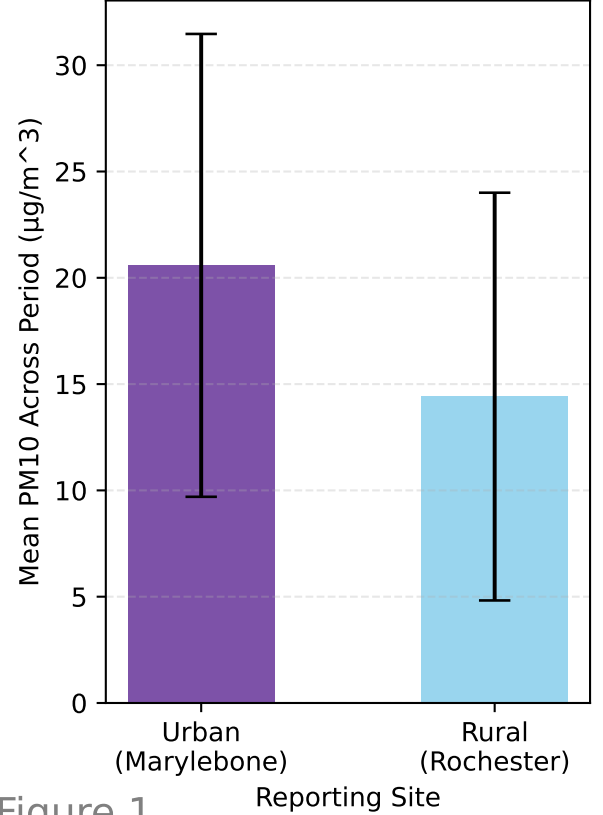


Figure 1

FIG. 1. Mean PM10 concentration across the full study period (January 2018 - February 2021) at the urban Marylebone Road and rural Rochester Stoke monitoring sites.

of approximately  $21 \mu\text{g}/\text{m}^3$  against a summer baseline of around  $12 \mu\text{g}/\text{m}^3$ , consistent with tillage-induced dust emission from dry spring soils [3]. The weaker April elevation in the Marylebone results may reflect the same transported continental agricultural dust superimposed on its higher traffic baseline [4], suggesting that the April signal is partly shared between sites rather than exclusively rural in origin.

To confirm that the April peak at Rochester represents a recurring seasonal signal rather than a single anomalous year, monthly PM10 was plotted separately for each year of the study period as shown in Fig. 3. The April peak is visible at Rochester in 2019 and 2020 (as well as a May peak seemingly delayed in 2018), confirming it as seasonal rather than anomalous. Notably, 2020 shows a reduced PM10 reading across both sites, consistent with the lower number of commuters due to Covid-19 restrictions.

To investigate shorter term, specifically diurnal structures in PM10 readings consistent with commuter traffic, mean PM10 was plotted by hour of day for both sites in

Mean Monthly PM10 Climatology (2018-2021)

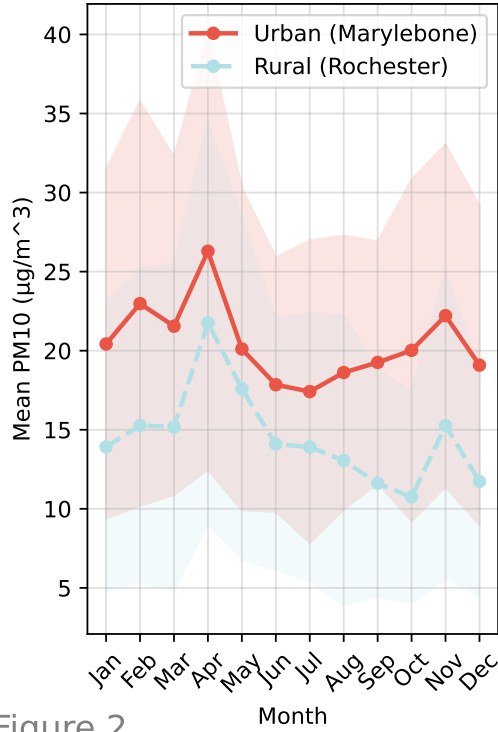


Figure 2

FIG. 2. Mean monthly PM10 climatology for urban (Marylebone) and rural (Rochester) sites, averaged across 2018-2021. Shaded regions indicate one standard deviation.

Fig. 4. Marylebone shows a clear morning peak between 8.00 and 10.00 and a secondary evening elevation between 17.00 and 21.00, coinciding with the morning and evening commute for most workers respectively. Rochester however shows no corresponding diurnal structure, remaining broadly flat throughout the day. Upon deeper inspection though it became clear that plotting the gradients for each mean hour may prove fruitful in evidence surrounding the hypothesis. The gradient analysis in Fig. 5 quantifies this further: Marylebone shows strongly positive gradients from 5.00 onwards, peaking at 0.77 with a secondary positive gradient during the evening commute window. Rochester gradients remain near zero throughout both commute windows, confirming the absence of any traffic driven diurnal signals at the rural site. The contrast between sites is consistent with the mechanical resuspension of road dust by commuter traffic at Marylebone [2], with UK commuters typically travelling during these windows [10]

To test whether the diurnal pattern at Marylebone is driven specifically by commuter traffic rather than a generic effect, hourly PM10 was assigned to weekday or weekend in Fig. 6. On weekdays, Marylebone shows a noticeable morning peak consistent with the morning commute, which is visibly weaker on weekends when the same roads carry significantly less commuter traffic. The

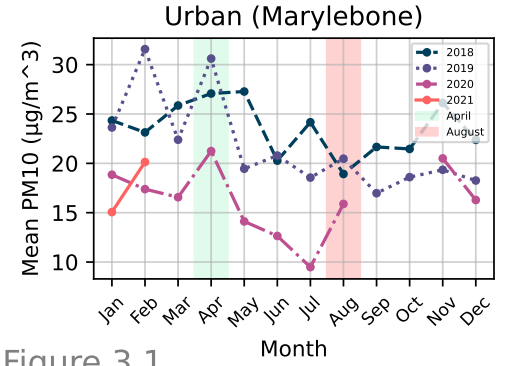


Figure 3.1

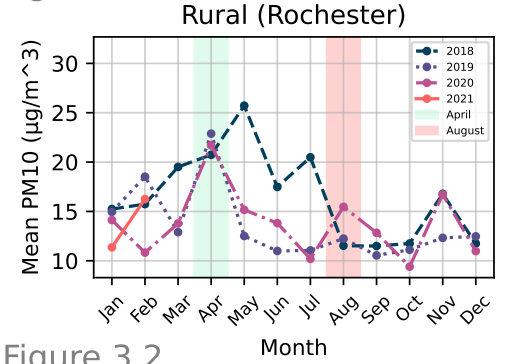


Figure 3.2

FIG. 3. Year-by-year monthly mean PM10 for urban Marylebone (top) and rural Rochester (bottom) across 2018-2021. Green and red shading highlight April and August respectively.

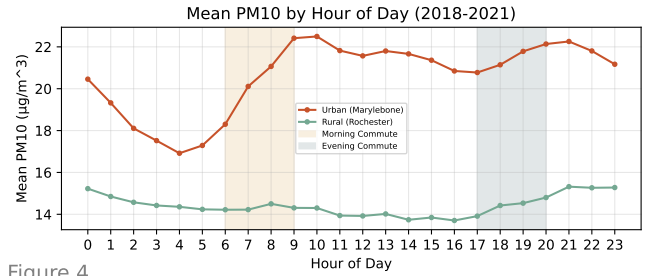


Figure 4

FIG. 4. Mean PM10 by hour of day (2018-2021) at urban Marylebone and rural Rochester. Orange and blue shading indicate morning and evening commute windows respectively.

weekend profile instead shows a high overnight PM10 which begins declining throughout the morning, suggesting a source outside working hours. Rochester shows no meaningful weekday/weekend adherence at any hour, confirming that the rural site is unaffected by urban activity cycle. The presence of the weekday morning peak and its absence at weekends constitutes strong evidence that commuter traffic is the primary driver of the diurnal PM10 structure at Marylebone [2].

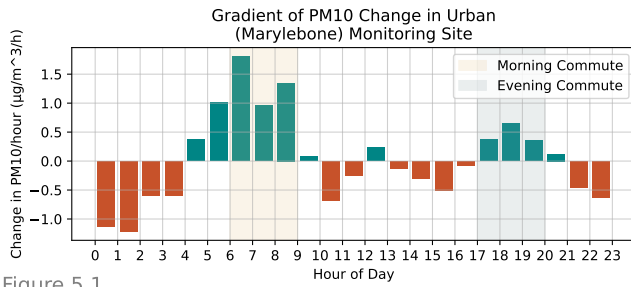


Figure 5.1

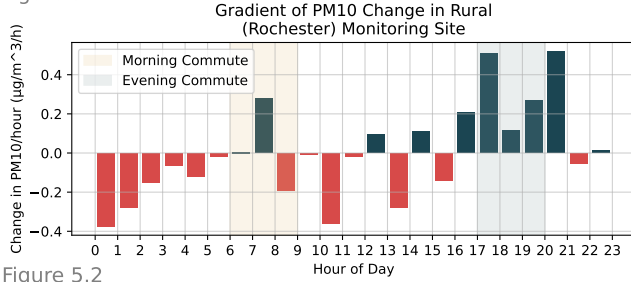


Figure 5.2

FIG. 5. Hourly gradient of PM10 change ( $\mu\text{g}/\text{m}^3/\text{h}$ ) at urban Marylebone (top) and rural Rochester (bottom). Positive bars indicate rising PM10, negative bars falling. Commute windows shaded as in Fig. 4.

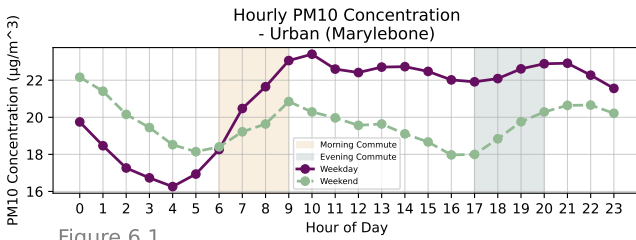


Figure 6.1

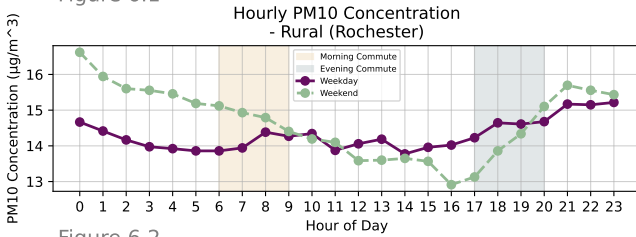


Figure 6.2

FIG. 6. Weekday vs weekend diurnal PM10 concentration at urban Marylebone (top) and rural Rochester (bottom), averaged across 2018-2021.

## B. Part 2: Meteorological Drivers of PM

To investigate which meteorological conditions drive PM10 variability at the rural Rochester site, Spearman rank correlations were calculated between PM10 and each of six weather variables across the 3 year study period. The overall results are presented in Table I. Five of the six variables showed statistically very significant associations with PM10 ( $p < 0.001$ ). Wind gust and wind speed show the strongest negative correlations, which

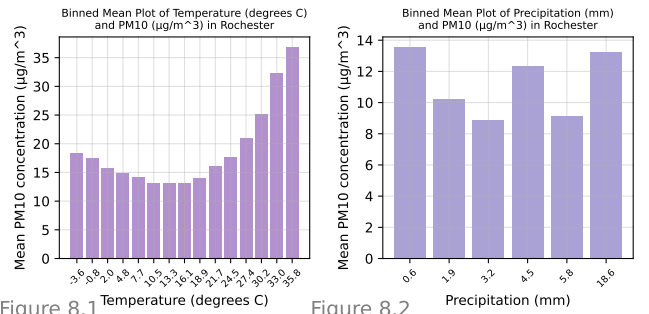


Figure 8.1

Figure 8.2

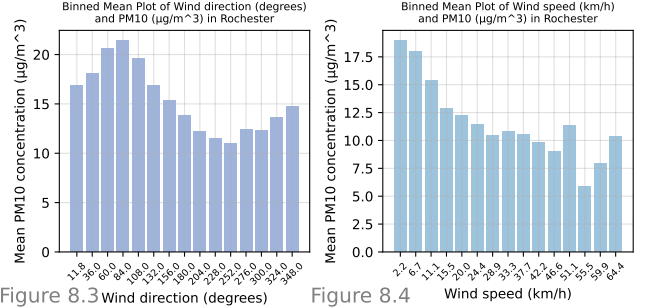


Figure 8.3

Figure 8.4

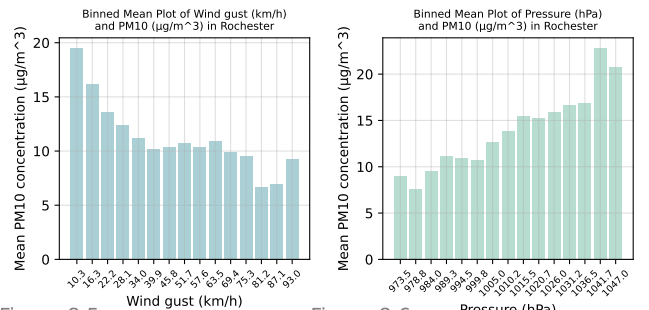


Figure 8.5

Figure 8.6

FIG. 7. Binned mean PM10 concentration against six meteorological variables at Rochester: temperature, precipitation, wind direction, wind speed, wind gust, and pressure (2018-2021).

lines up with the hypothesis that atmospheric dispersal by high winds reduces PM10 readings. Pressure shows the strongest positive correlation of  $\rho = 0.223$ . This is also consistent with the proposed theory of anticyclones having a role in PM10 particulate distribution: Stronger pressures create anticyclones better at trapping particulates at an observable level. [5, 11]. Temperature however shows no significant overall association.

The binned mean plots (Fig. 7) are however a useful tool in observing these relationships. As an example, the temperature plot which was statistically shown to have no inherent association clearly correlates with a high level of PM10 concentration this may be caused by a "threshold" point being reached where the mechanism which correlates with PM10 rises begins. Because the temperature is mostly moderate, it may rarely reach this point. Pressure too shows a progressive increase above 1000 hPa, consistent with the anticyclonic hypothesis, also precipitation showing a strong non linear response as after a small amount of rain the PM10 seems to dip

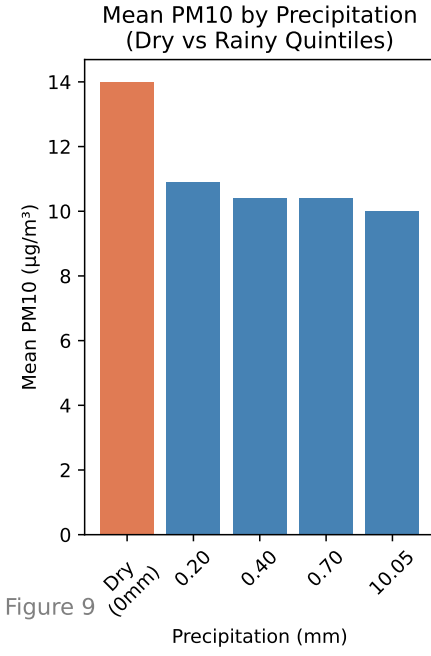


Figure 9

FIG. 8. Mean PM10 concentration by precipitation quintile at Rochester (2018-2021), separating dry conditions (0 mm) from rainy quintiles.

massively. The dry quantile of precipitation when plotted as it's own bin is clearly linked to a higher PM10 observation, as shown in Fig. 8. This threshold effect suggests that wet deposition is highly efficient at removing particulates, even at low rain intensity.

Now that it's established that different weather circumstances can change the PM10 levels, it's logical to find the Spearman's Rank Correlation between each season's PM10 levels and the categories. Table III reveals how different the temperature for one truly behaves across seasonal boundaries. The seasonal Spearman's Rank coefficients are significant in support of the soil moisture hypothesis, as they align with its predictions of temperature and the newlyfound observations for pressure. For temperature, the change of  $\rho$  by over 0.5 between Winter and Summer points to the threshold mechanism discussed before. During Winter, it never gets warm enough to activate, even sometimes higher temperatures melting ice, increasing soil moisture. During Summer however, the threshold can be consistently reached, allowing the temperature to reduce the soil moisture and leading to a positive correlation between temperature and PM10 measurements [3]. Similarly for pressure, because of the theory of anticyclones causing lower wind speeds and trapping PM10 particles near the surface, the hypothesis can explain why the values of  $\rho$  change across seasons. In Winter, anticyclones are more effective at trapping PM10 at observable layers [11], hence the higher observed values. Furthermore, pressure even shows its strongest positive value for  $\rho$  in Winter, (namely +0.363), which further supports the anticyclonic

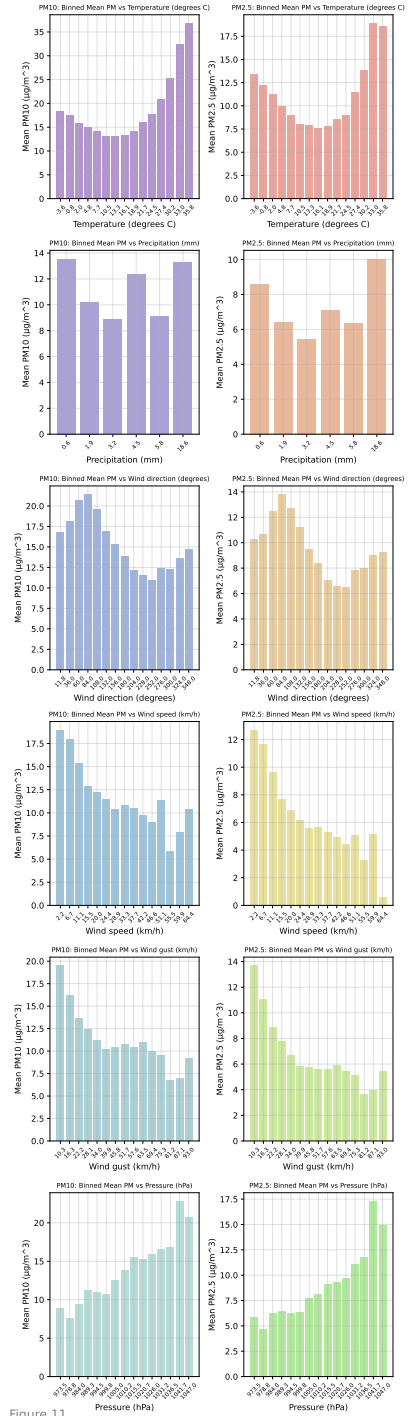


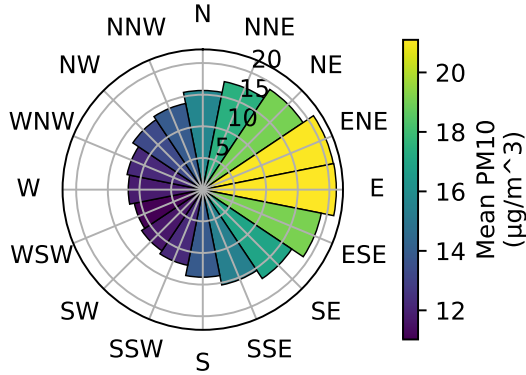
Figure 11

FIG. 9. Side-by-side binned mean plots comparing PM10 (left) and PM2.5 (right) against all six meteorological variables at Rochester (2018-2021).

theory as it is predominantly a cold season phenomenon. [11]

One avenue not explored yet is the direction of wind. A windrose can be used to visually plot this (Fig. 10). The wind rose is a strikingly clear directional signal: mean PM10 observation is highest under easterly and north-

Mean PM10 per Wind Direction  
Rural (Rochester, 2018-21)



Mean PM2.5 per Wind Direction  
Rural (Rochester, 2018-21)

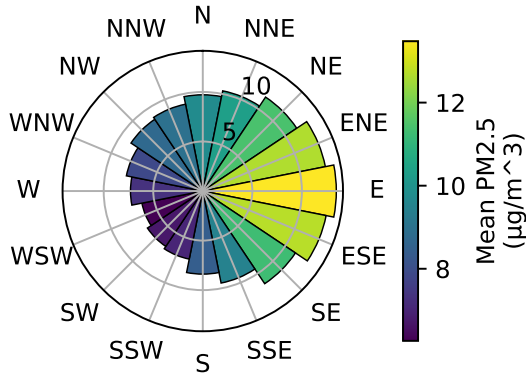


Figure 12

FIG. 10. Comparative wind roses showing mean PM10 (left) and PM2.5 (right) per wind direction at rural Rochester (2018-2021).

easterly winds, reaching  $20 \mu\text{g}/\text{m}^3$  compared to its baseline of approximately  $12 \mu\text{g}/\text{m}^3$  under opposite westerly winds. This is consistent with long range transport of continental European pollution and agricultural dust, as the UK (and in particular Rochester) is West of not only the Netherlands and Germany but also the English Channel, all sources of significant PM10 and PM2.5 particles (as the comparison wind rose shows) [4].

Comparing the values of PM10 versus PM2.5 now, the idea of a Spearman test Rho comparison seems insightful. If you consider the values in Table II, apart from temperature which was discussed separately, all of the Rho values are surprisingly similar to each other. Pressure has  $+0.223$  to  $+0.233$  for PM10 and PM2.5, Pre-

TABLE I. Spearman rank correlation coefficients ( $\rho$ ) between meteorological variables and PM10 at Rochester (2018–2021). Significance threshold:  $p < 0.05$ .

| Variable                           | $\rho$   | $p$ -value | Significant? |
|------------------------------------|----------|------------|--------------|
| Temperature ( $^{\circ}\text{C}$ ) | $-0.005$ | $0.387$    | No           |
| Precipitation (mm)                 | $-0.159$ | $< 0.001$  | Yes          |
| Wind direction ( $^{\circ}$ )      | $-0.251$ | $< 0.001$  | Yes          |
| Wind speed (km/h)                  | $-0.288$ | $< 0.001$  | Yes          |
| Wind gust (km/h)                   | $-0.298$ | $< 0.001$  | Yes          |
| Pressure (hPa)                     | $+0.223$ | $< 0.001$  | Yes          |

TABLE II. Spearman rank correlation coefficients ( $\rho$ ) comparing meteorological drivers of PM10 and PM2.5 at Rochester (2018–2021).

| Variable                           | $\rho$ (PM10) | $p$ (PM10) | $\rho$ (PM2.5) | $p$ (PM2.5) |
|------------------------------------|---------------|------------|----------------|-------------|
| Temperature ( $^{\circ}\text{C}$ ) | $-0.005$      | $0.387$    | $-0.088$       | $< 0.001$   |
| Precipitation (mm)                 | $-0.159$      | $< 0.001$  | $-0.146$       | $< 0.001$   |
| Wind direction ( $^{\circ}$ )      | $-0.251$      | $< 0.001$  | $-0.213$       | $< 0.001$   |
| Wind speed (km/h)                  | $-0.288$      | $< 0.001$  | $-0.357$       | $< 0.001$   |
| Wind gust (km/h)                   | $-0.298$      | $< 0.001$  | $-0.380$       | $< 0.001$   |
| Pressure (hPa)                     | $+0.223$      | $< 0.001$  | $+0.233$       | $< 0.001$   |

cipitation has  $-0.159$  to  $-0.146$  for PM10 and PM2.5 and this pattern follows for all 5 values. However, when considering more carefully, it's noticeable that for Wind speed and wind gust values are  $-0.288$  to  $-0.357$  and  $-0.298$  to  $-0.380$  for PM10 and PM2.5 all respectively. This suggests that finer particles are more dispersed by the same winds. When further considering Fig. 9 it becomes clear that the two particle families have a very close relationship. They share almost identical binned mean plots, supporting the hypothesis that they'll respond overwhelmingly similar to their environmental factors due to their very similar relationships to them.

#### IV. CONCLUSION

This report investigated two questions: whether PM10 differs systematically between urban and rural sites, and whether meteorological conditions are significant drivers of PM variability at the rural site. The urban Marylebone site recorded significantly higher mean PM10 than rural Rochester across the study period ( $20.6$  vs  $14.4 \mu\text{g}/\text{m}^3$ , Mann-Whitney  $p < 0.001$ ,  $r = 0.331$ ), confirming the first hypothesis. The diurnal and week-day/weekend analysis demonstrates that this elevation is driven primarily by commuters and their traffic, with the urban morning peak absent at weekends and showing no equivalent at the rural site. Rochester's pronounced April PM10 peak is consistent with tillage-induced agricultural dust emission from dry spring soils [3].

Meteorological conditions at Rochester show clear and statistically significant associations with PM10. High



TABLE III. Seasonal Spearman rank correlation coefficients ( $\rho$ ) for PM10 (top) and PM2.5 (bottom) against meteorological variables at Rochester (2018–2021).

| <b>PM10</b>        |        |        |        |        |
|--------------------|--------|--------|--------|--------|
| Variable           | Winter | Spring | Summer | Autumn |
| Precipitation (mm) | −0.208 | −0.201 | −0.065 | −0.136 |
| Pressure (hPa)     | +0.363 | +0.162 | +0.100 | +0.133 |
| Temperature (°C)   | −0.201 | +0.160 | +0.311 | −0.104 |
| Wind direction (°) | −0.194 | −0.292 | −0.204 | −0.286 |
| Wind gust (km/h)   | −0.419 | −0.292 | −0.121 | −0.275 |
| Wind speed (km/h)  | −0.391 | −0.290 | −0.177 | −0.262 |
| <b>PM2.5</b>       |        |        |        |        |
| Variable           | Winter | Spring | Summer | Autumn |
| Precipitation (mm) | −0.210 | −0.197 | −0.048 | −0.125 |
| Pressure (hPa)     | +0.372 | +0.228 | +0.068 | +0.125 |
| Temperature (°C)   | −0.232 | +0.151 | +0.189 | −0.193 |
| Wind direction (°) | −0.158 | −0.239 | −0.218 | −0.237 |
| Wind gust (km/h)   | −0.495 | −0.401 | −0.241 | −0.351 |
| Wind speed (km/h)  | −0.455 | −0.362 | −0.258 | −0.336 |

wind speed and gusts disperse particulates, lowering readings, while high pressure anticyclonic conditions trap PM near the surface [5]. Easterly winds transport continental European pollution to the site [4], and even minimal precipitation sharply suppresses PM10 through wet deposition. The near-identical directional response of PM2.5 to all six meteorological variables confirms that both particle sizes share common meteorological controls.

A limitation of this study is that the weather data was sourced from Heathrow Airport rather than from either monitoring site directly, introducing potential representativeness error. Future work could incorporate site-specific meteorological measurements and extend the analysis to include gaseous pollutants such as NO<sub>2</sub> to more fully characterise urban air quality drivers.

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